Subject: Mathematics **WEEKLY CHAPTER TEST – 04 (QUADRATIC EQUATIONS)** **Time:** 45 Minutes

Class: Class X (CBSE) **Chapter 4: Quadratic Equations** **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choice is provided in Question 10(iii).
2. **Section A** comprises 4 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 1 VSAQ of 2 marks and 3 SA questions of 3 marks each.
4. **Section C** comprises 1 Applied Case Study with 3 sub-parts of 4 marks total.

SECTION A: Multiple Choice Questions (5 × 1 = 5 Marks)

Q1. The value of c for which the equation $ax^2 + 2bx + c = 0$ has equal roots is: [1]

(A) $\frac{b^2}{a}$ (B) $\frac{b^2}{4a}$ (C) $\frac{a^2}{b}$ (D) $\frac{a}{b^2}$

Q2. If the equation $9x^2 + 6kx + 4 = 0$ has equal roots, then the roots are both equal to: [1]

(A) $\pm\frac{2}{3}$ (B) $\pm\frac{3}{2}$ (C) 0 (D) ± 3

Q3. If the roots of the quadratic equation $ax^2 + bx + c = 0$ are equal, then $c =$: [1]

(A) $-\frac{b}{2a}$ (B) $\frac{b}{2a}$ (C) $-\frac{b^2}{4a}$ (D) $\frac{b^2}{4a}$

Q4. The positive value of k for which both $x^2 + kx + 64 = 0$ and $x^2 - 8x + k = 0$ have real roots is: [1]

(A) 4 (B) 8 (C) 12 (D) 16

Q5. **Assertion (A):** If the roots of $(2k-1)x^2 + 4x - 3 = 0$ are reciprocal of each other, then $k = -1$.
Reason (R): If the roots of $ax^2 + bx + c = 0$ are reciprocal to each other, then $a = c$. [1]

(A) Both A and R are true and R is correct explanation of A. (B) Both are true, but R is not correct explanation.
(C) A is true, R is false. (D) A is false, R is true.

SECTION B: Short Answer Type Questions (Part I)

Q6. Find the value of k for which the quadratic equation $kx(x-2) + 6 = 0$ has two equal roots. [2]

Q7. Find the values of k for which the equation $(k-12)x^2 + 2(k-12)x + 2 = 0$ has real and equal roots. [3]

SECTION B: Short Answer Type Questions (Continued) (Total Sec B = 11 Marks)

Q8. If the roots of the equation $(c^2 - ab)x^2 - 2(a^2 - bc)x + (b^2 - ac) = 0$ are equal, prove that either $a = 0$ or $a^3 + b^3 + c^3 = 3abc$. [3]

Q9. Find the values of p for which the quadratic equation $(2p + 1)x^2 - (7p + 2)x + (7p - 3) = 0$ has equal roots. Also find the roots. [3]

SECTION C: Applied Case Study Based Question (1 × 4 = 4 Marks)

Q10. Case Study: Petroleum Industry & Crude Oil Production Economics

An oil refining company extracts crude oil in barrels. Due to processing limits, maximum production is 300 barrels per day. Financial economists determined that the net daily profit $P(x)$ (in rupees) from selling x barrels is given by:

$$P(x) = -10x^2 + 3500x - 66000$$

Based on the above financial model, answer the following questions:

(i) If the company operates at break-even ($P(x) = 0$), write the simplified quadratic equation in standard form. [1]

(ii) Calculate the discriminant of the break-even quadratic equation. [1]

(iii) Determine the minimum number of barrels the company must produce to break even. [2]

OR

Find the production level x that maximizes the daily profit, and calculate that maximum profit. [2]

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